



IDC216 (Statistics for Scientists) / Monsoon 2025 / Mid Semester Exam 1
Maximum Marks : 15

September 02, 2025

- All questions will be considered in the evaluation.
- Use of calculators is allowed. However, exchange of calculators and use of mobile apps is not permitted.
- Organize your work, in a reasonably neat and coherent way. Work scattered all over the pages without a clear ordering may be missed by the evaluator. Illegible work will not receive credit.
- Mysterious or unsupported answers will not receive credit. A correct answer, unsupported by calculations, explanation, or algebraic work will receive no credit; an incorrect answer supported by substantially correct calculations and explanations *might* still receive partial credit.

1. The following data captures the change in the temperature of the test-tube where a chemical reaction is being performed: [3+2]

Time (min)	0	5	10	15
Temperature °C	25	35	48	65

- (a) Use Newton's divided difference method to estimate the temperature at $t = 12$ minutes.
- (b) Comment briefly: Would you use the same method to predict the temperature at $t = 40$ minutes? Why?
2. A study measures the relation between light intensity x (lux) and the rate of photosynthesis y (mg of O_2 released/hour). The researcher tabulates (x, y) data as follows: [3+2]
- (20, 5), (40, 8), (60, 16), (80, 19), (100, 24)
- (a) Fit a least-squares regression line $y = b_0 + b_1x$.
- (b) Find the coefficient of determination R^2 for this regression line.
3. The dataset of annual rainfall (mm) over 10 consecutive years in a city is as follows: [3+2]
- {820, 790, 815, 830, 1200, 810, 805, 795, 820, 825}.
- (a) Compute *signal-to-noise ratio* of this data.
- (b) Comment on the skewness of this data.